

Big Mart Sales Analysis Internship Project Report

Project Title - **Big Mart Sales Analysis Using Power BI**

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Introduction

The Big Mart Sales Analysis project focuses on analyzing retail data using Power BI. The project is designed to understand product-level and outlet-level information in a structured way and present meaningful insights through interactive dashboards.

Retail businesses generate large amounts of data related to products, pricing, outlet types, product visibility, and item categories. Analyzing this data helps identify patterns that can support better decision-making. Power BI is used in this project to clean, transform, analyze, and visualize the dataset in a professional and user-friendly format.

This internship project provides practical exposure to business intelligence concepts, data preparation, dashboard development, and visual storytelling.

Objectives

The main objectives of the Big Mart Sales Analysis project are:

- To understand the structure and features of the Big Mart dataset
- To prepare and clean the data for accurate analysis
- To analyze product-related and outlet-related information
- To study product categories, item visibility, pricing, and outlet characteristics
- To create interactive reports and dashboards using Power BI
- To present insights in a clean, simple, and professional format
- To improve practical skills in data analysis and business intelligence

Dataset Description

The dataset contains information about products and outlets of Big Mart. It includes item-level details, pricing information, product visibility, and outlet characteristics.

Main Dataset Fields

Column Name	Description
Item Identifier	Unique identification code for each product
Item Fat Content	Indicates whether the product is low fat, regular, or belongs to another fat content category
Item MRP	Maximum Retail Price of the product
Item Type	Category of the product, such as food, household, dairy, frozen foods, or soft drinks
Item Visibility	Percentage of total display area allocated to the product in the outlet
Item Weight	Weight of the product
Outlet Identifier	Unique identification code for each outlet
Outlet Establishment Year	Year in which the outlet was established
Outlet Size	Size of the outlet, such as small, medium, or high
Outlet Location Type	Location category of the outlet, such as Tier 1, Tier 2, or Tier 3
Outlet Type	Type of outlet, such as supermarket or grocery store

Methodology

The project followed a structured data analysis process using Power BI. The methodology included data understanding, data cleaning, transformation, analysis, visualization, and dashboard creation.

1. Data Understanding

The dataset was reviewed to understand the available columns, data types, and business meaning of each field. Product-related and outlet-related variables were identified for analysis.

Key focus areas included:

- Product identification
- Product category
- Fat content
- Item price
- Product visibility
- Outlet size
- Outlet location
- Outlet type

2. Data Preparation and Cleaning

Data preparation was performed to improve data quality and make the dataset suitable for analysis.

Data cleaning activities included:

- Checking missing values in important columns
- Reviewing inconsistent entries in categorical fields
- Standardizing item fat content values

- Checking data types for numerical and categorical columns
- Removing duplicate or unnecessary records where applicable
- Renaming columns for better readability
- Preparing clean fields for dashboard visuals

3. Data Transformation

Power Query was used to transform the raw dataset into a structured analytical format. Data transformation helped in improving consistency and usability.

Transformation steps included:

- Formatting column names
- Correcting inconsistent category values
- Handling missing or blank values
- Converting data types
- Creating calculated fields where required
- Organizing product and outlet details for visualization

4. Data Analysis

After cleaning and transformation, the dataset was analyzed to identify useful insights. The analysis focused on product performance indicators, outlet characteristics, pricing patterns, and visibility distribution.

5. Dashboard Development

Interactive dashboards were created in Power BI using different visualization elements. The dashboard was designed to be simple, clean, and easy to understand.

Power BI Tools Used

The following Power BI features and tools were used in the project:

Tool/Feature	Purpose
Power Query	Data cleaning, transformation, and preparation
Data Model	Organizing relationships and fields for analysis
Visualizations	Creating charts, graphs, and summary views
KPI Cards	Displaying important numerical indicators
Slicers	Adding interactive filters for better exploration
Bar Charts	Comparing product categories and outlet details
Pie/Donut Charts	Showing category-wise distribution
Tables	Presenting detailed data in structured format
Formatting Tools	Improving dashboard appearance and readability

Key Analysis Areas

1. Product Categories

Product categories were analyzed to understand the distribution of different item types available in Big Mart outlets.

Analysis included:

- Count of products by item type
- Comparison of different product categories
- Identification of major and minor product groups
- Category-wise product visibility and pricing patterns

2. Outlet Characteristics

Outlet-level analysis was performed to understand how outlet size, location, and type are distributed.

Analysis included:

- Outlet count by outlet type
- Outlet distribution by location type
- Outlet size comparison
- Establishment year overview
- Relationship between outlet features and product availability

3. Pricing Analysis

Item MRP was analyzed to study pricing patterns across different products and categories.

Analysis included:

- Average MRP by item type
- Highest and lowest priced product categories
- Price variation across different outlet types
- Product pricing comparison through visuals

4. Product Visibility

Item visibility was analyzed to understand how much display space is given to products in outlets.

Analysis included:

- Visibility by item type
- Products with high and low visibility
- Relationship between visibility and product category
- Outlet-wise visibility trends

5. Fat Content Analysis

The item fat content field was analyzed to compare product types based on fat content.

Analysis included:

- Count of products by fat content
- Distribution of low fat and regular products
- Product category comparison based on fat content

Dashboard Features

The Power BI dashboard was created with a professional layout and interactive elements.

Main Dashboard Features

- Interactive filters for item type, outlet type, outlet size, and location type
- KPI cards for quick summary indicators
- Clean visual layout for easy interpretation
- Consistent color theme and formatting
- Clear titles and labels for each visual
- Category-wise and outlet-wise comparison charts
- Simple navigation and user-friendly dashboard structure
- Professional presentation suitable for internship project submission

Dashboard Design Approach

The dashboard was designed using a clean and structured approach. The main focus was on readability, simplicity, and effective presentation of insights.

Design principles followed:

- Minimal and professional layout
- Consistent colors across visuals
- Clear chart titles
- Proper spacing between visuals
- Easy-to-read fonts
- Highlighting important metrics through KPI cards
- Use of slicers for interactive analysis

Skills Learned

This internship project helped in developing practical data analytics and business intelligence skills.

Key Skills Gained

- Data cleaning using Power Query
- Data transformation in Power BI
- Understanding retail business data
- Creating interactive dashboards
- Designing professional reports
- Using KPI cards and slicers
- Building meaningful visualizations
- Analyzing product and outlet-level data
- Improving data storytelling skills
- Applying business intelligence concepts
- Presenting insights in a structured format

Challenges Faced

During the project, several challenges were encountered while preparing and analyzing the data.

Main Challenges

- Understanding the structure of the retail dataset
- Identifying important columns for analysis
- Handling missing values in item weight and outlet size
- Standardizing inconsistent category names
- Selecting suitable visuals for each analysis area
- Designing a dashboard that is simple and informative
- Maintaining consistent formatting across all visuals
- Presenting insights clearly without making the dashboard overcrowded

Solutions Applied

The challenges were handled through a systematic approach.

Actions Taken

- Dataset columns were studied carefully before analysis
- Power Query was used for cleaning and transformation
- Inconsistent category values were corrected
- Missing values were reviewed and handled carefully
- Visuals were selected based on the type of data
- Dashboard layout was planned before final design
- KPI cards and slicers were used to improve interactivity
- Unnecessary fields were avoided to maintain clarity

Future Scope

The Big Mart Sales Analysis project can be improved further by adding more advanced analysis features.

Possible Future Enhancements

- Adding actual sales data for deeper performance analysis
- Performing time-based analysis using monthly or yearly sales data
- Creating sales trend reports
- Adding forecasting models for future sales prediction
- Comparing outlet performance over different time periods
- Analyzing customer purchase behavior
- Using advanced DAX measures for detailed calculations
- Adding profit, revenue, and target-based KPIs
- Creating separate dashboards for product, outlet, and sales performance
- Including predictive analytics for business decision-making

Conclusion

The Big Mart Sales Analysis internship project provided practical experience in analyzing retail data using Power BI. The project covered important stages of data analytics, including data understanding, data cleaning, transformation, visualization, and dashboard development.

Through this project, Santoshi Pandalwad gained hands-on experience in preparing retail data, creating interactive Power BI reports, and presenting insights in a professional format. The analysis of product categories, outlet characteristics, pricing, visibility, and fat content helped in understanding how retail data can be used for business intelligence.

Overall, the project strengthened skills in data cleaning, visualization, dashboard design, and analytical thinking. It also provided valuable exposure to real-world data analysis practices and demonstrated the importance of Power BI in converting raw data into meaningful business insights.